

ALIVE TILL NIGHTFALL

Development of a 3D Asymmetric Tag Game

A. Boavida, P. Schlup, J. Erik

University of the Algarve | Masters in Informatics Engineering

CONTENTS

1. Introduction & Mechanics
 - Gameplay Loop
2. Technical Architecture
3. Artificial Intelligence
 - NavMesh & Behaviors
4. Level Design & UI

INTRODUCTION & MECHANICS

- **Core Concept:** 3D asymmetric competitive tag game developed in Unity.
- **Roles:** **Hunter** (chases) vs. **Runner** (evades).
- **Match Structure:**
 - Best-of-5 rounds format.
 - 30-second time limit per round.
 - Roles alternate between Human and Machine every round.

Ultimate Objective

The first entity to achieve 3 victories is declared the winner of the match.

TECHNICAL ARCHITECTURE

- **Centralized State:** The `#stress[GameManager]` uses a Singleton pattern to control round transitions, scoring, and UI states.
- **State Machine Pattern:** - The `GameCharacter` script acts as the physical entity.
 - It dynamically delegates logic to `HumanPlayerController` or `MachinePlayerController`.
- **Physics-Based Locomotion:**
 - Uses Unity's New Input System.
 - Movement relies on `Rigidbody.MovePosition` and `MoveRotation` to prevent clipping geometry.

- Custom CameraFollow script manages viewport tracking.
- **Smooth Tracking:** Uses `Vector3.Lerp` to trail the active human player smoothly.
- **Occlusion Prevention:** Utilizes `Physics.SphereCast` to detect walls and pull the camera forward.
- **Map Transitions:** `#stress[SnapToTarget()]` method instantly teleports the camera to prevent jarring visual panning when swapping arenas.

- Downloaded FBX models (Goblin, Knight) already contained embedded clips: Idle, Run, Walk (+ _Carry variants)
- Added an Animator Controller per model with two states, wired via a single float parameter Speed
- Transitions: Idle → Run when Speed > 0.1, Run → Idle when Speed < 0.1 — Has Exit Time disabled for instant response
- GameCharacter drives Speed every frame from the active IPlayerController:
 - Human: derived directly from **input**, not physics velocity (instant, no lag from Rigidbody damping)
 - AI: derived from NavMeshAgent.velocity (correctly shows Idle when physically blocked, Run once actually moving)

- Built on Unity's built-in **procedural skybox** shader — no external asset needed
- SkyboxDayNightController clones the skybox material at runtime (never mutates the shared asset)
- Lerps `_SkyTint`, `_GroundColor`, `_Exposure`, `_AtmosphereThickness` from warm evening tones to dark navy night
- Optionally dims/reddens the directional sun light alongside it

- Transition duration = **max round length (30s)**, restarted every round via `GameManager.StartRound()`
- Creates a subtle “time pressure” cue — the world visibly darkens as the round clock runs out

- New TitleScreenCameraPan script orbits slowly around a fixed look-target placed in Arena 3
- Runs continuously while the start screen is shown, giving a cinematic living background instead of a static menu
- GameManager swaps cameras automatically:
 - ShowStartScreen() → title camera on, gameplay camera off, **all arenas reactivated** (so any arena can be shown as backdrop)
 - StartGame() → gameplay camera on, title camera off
- Avoided parenting the camera under an arena object — arenas get toggled on/off per round, which would've disabled the camera along with it

ARTIFICIAL INTELLIGENCE

Hunter AI

- Aggressive pathfinding loop.
- Continuously updates NavMeshAgent destination directly to the Runner's coordinates.

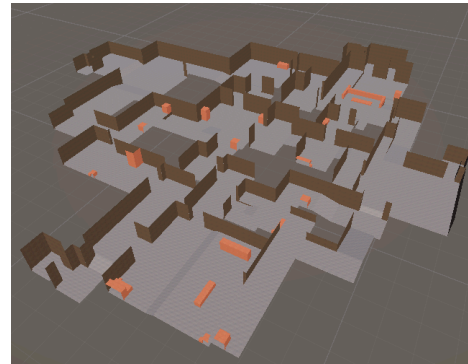
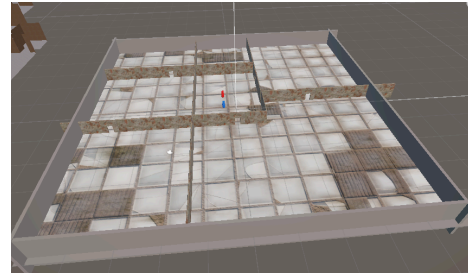
Runner AI

- Dynamic evasion heuristics.
- Queries `Physics.OverlapSphere` to find obstacles.
- Selects candidate hiding spots on the opposite side of obstacles relative to the Hunter.

- The AI temporarily sets the Rigidbody to “Kinematic” to allow the NavMeshAgent full spatial control without physical conflicts.

LEVEL DESIGN & UI

- The GameManager dynamically loads maps based on the round.
- **Arena 1:** Baseline mechanics and simple geometry.
- **Arena 2:** Complex elevations requiring baked NavMesh Surfaces.
- **Arena 3:** The final tiebreaker showdown.



Custom invisible physics boundaries prevent characters from falling out of bounds.

- Responsive Canvas utilizes Scale With Screen Size for multi-resolution support.
- **Interface States:**
 1. **Start & Instructions:** Pre-game initialization and rules.
 2. **HUD:** Real-time round number, score, and dynamic timer.
 3. **End Screen:** Final victor declaration and computed session statistics.



Thank you!

Any questions?